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Mini DC UPS with Op-Amp Based Analog Deep-Discharge Protection

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Abstract - This paper presents the design, mathematical modeling, and Proteus-based simulation of a fully analog, solid-state 3S Lithium-Ion DC Uninterruptible Power Supply (UPS) intended to keep a Wi-Fi router (TP-Link Archer C54, 9 V, 0.85 A) continuously powered during mains outages while protecting the battery pack from irreversible deep-discharge damage. Protection logic is realized without any microcontroller, using a discrete comparator built around an LM324 operational amplifier, a 1N4733A Zener voltage reference, a BC547 driver transistor, and an IRFZ44N power MOSFET load switch. Passive components were analytically sized to enforce a hard voltage cutoff at 9.6 V, reserving 10–15% of the pack capacity to extend cycle life, while an LM2596 buck converter regulates the output to a stable 9 V rail. Transient simulation of a 100 s grid-outage scenario shows a seamless transfer to battery power at the moment of outage, a 24.3s controlled discharge window, and a hard disconnection of the load current from approximately 1.05 A to near 0 A once the 9.6 V threshold is reached, confirming that the proposed analog protection scheme reliably prevents over-discharge without any programmable logic.

Keywords - DC uninterruptible power supply; Lithium-ion battery protection; deep-discharge cutoff; operational-amplifier comparator; power MOSFET switch; buck converter.

I. INTRODUCTION

Network equipment such as home and small-office routers must remain continuously energized to preserve connectivity, yet grid load-shedding and unplanned outages are common in many regions. A DC uninterruptible power supply (UPS) bridges this gap by seamlessly transferring the load to a battery pack the instant mains power is lost. Lithium-ion (Li-

ion) cells are an attractive energy-storage medium for this purpose because of their high energy density and low self-discharge rate compared with lead-acid batteries; however, unlike lead-acid chemistries, Li-ion cells suffer permanent capacity loss and, in extreme cases, internal short-circuiting if they are discharged below a safe cell voltage. A dependable UPS for Li-ion storage must therefore combine (i) millisecond-scale switchover to battery power, (ii) regulated output for the protected load, and (iii) a hardware-enforced low-voltage disconnect that removes the load before the pack is driven into an unsafe state.

This project develops a compact 3S (11.1 V nominal) Li-ion UPS for a TP-Link Archer C54 router. The design deliberately avoids a microcontroller-based battery management system (BMS) in favor of a fully analog comparator circuit, so that the protection function is deterministic, immune to firmware faults, and inexpensive to reproduce on a single-sided prototype board.

II. RELATED WORK

Battery management systems for Li-ion packs are typically implemented either as dedicated analog supervisor circuits or as microcontroller-supervised digital systems. Cabrera et al. describe a reconfigurable analog BMS in which comparator ICs continuously monitor per-cell voltage and disconnect a parallel MOSFET bank once any cell falls below a minimum threshold, demonstrating that a purely analog centralized topology can protect a multi-cell pack without a digital controller [1]. Separately, Maden and Arabacı show experimentally that the choice of discharge cut-off voltage directly trades off usable capacity against long-term cycle life, and that terminating discharge before the manufacturer-specified deep-discharge point measurably extends service life at the cost of some

available energy [2]. These two findings motivate the present design: an analog, deterministic cutoff (following [1]) set conservatively above the cell's absolute minimum voltage (following the capacity/lifetime trade-off in [2]).

III. PROBLEM STATEMENT

Three recurring failure modes were identified as the design targets for this project:

- 1) Frequent mains interruptions cause repeated network drops for equipment with no local backup power.
- 2) Lead-acid batteries, the conventional low-cost UPS storage medium, are inefficient and degrade quickly under the shallow, frequent cycling typical of load-shedding.
- 3) Li-ion packs used as drop-in replacements are frequently damaged because consumer UPS circuits lack an automatic, fail-safe low-voltage cutoff, permitting the cells to be driven into a deep-discharge state.

IV. PROPOSED SYSTEM ARCHITECTURE

The proposed UPS is organized into four cooperating stages, shown schematically in Fig. 1 and summarized by component in Table I.

A. Power and Regulation Stage

A 12.6 V constant-voltage adapter float-charges the 3S battery pack (B1) in parallel with the load path, so that transfer to battery power on grid loss is instantaneous and glitch-free. An LM2596 buck converter subsequently steps the battery-rail voltage down to a regulated 9 V output for the router.

B. Voltage Reference Stage

A 1N4733A Zener diode (D1), biased through R1, establishes a constant 5.1 V reference rail that is independent of the battery's state of charge and is used as the comparator's inverting-input threshold.

C. Comparator Logic Stage

One section (U1:A) of an LM324 quad operational amplifier is configured as an open-loop comparator. It continuously compares a resistively scaled sample of the battery voltage, taken at the R2–RV1–R3 divider, against the fixed 5.1 V Zener reference.

D. Power Switching Stage

The comparator output drives a BC547 BJT (Q1), which in turn drives the gate of an IRFZ44N N-channel power MOSFET (Q2). The MOSFET switches the load's return (ground) path, and its low on-resistance $R_{DS(on)}$ keeps conduction losses negligible while the load is connected.

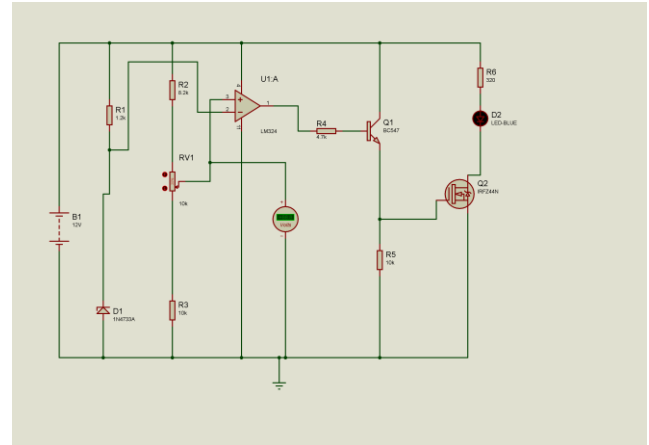


Fig. 1. Proteus schematic of the DC UPS showing the battery, Zener reference, LM324 comparator, and the BC547/IRFZ44N switching stage.

TABLE I
Component Designators and Values

Designator	Value / Part No.	Component	Function
R1	1.2 kΩ	Resistor	Zener bias setting
R2	8.2 kΩ	Resistor	Divider (coarse)
RV1	10 kΩ (≈623 Ω)	Trimpot	Divider (fine calib.)

Designator	Value / Part No.	Component	Function
R3	10 kΩ	Resistor	Divider (bottom leg)
R4	4.7 kΩ	Resistor	BJT base limiter
R5	10 kΩ	Resistor	MOSFET gate pull-down

Designator	Value / Part No.	Component	Function
R6	320 Ω	Resistor	LED current limiter
D1	1N4733 A	Zener diode	5.1 V reference
D2	LED (blue)	Indicator	Power-on status
Q1	BC547	NPN BJT	Gate driver
Q2	IRFZ44 N	N-MOSFET	Load ground switch

V. MATHEMATICAL MODELING AND CALIBRATION

All passive component values were analytically chosen to enforce a hard cutoff at exactly 9.6 V, which reserves approximately 10–15% of the pack's rated capacity as a safety margin and thereby extends the usable cycle life of the cells.

A. Zener Bias Current (R1)

The 1N4733A requires a minimum sustaining current to remain in regulation. With V_{rail} the battery-rail voltage and V_Z the 5.1 V Zener voltage,

$$I_Z = (V_{\text{rail}} - V_Z) / R_1 \quad (1)$$

At the 9.6 V cutoff threshold this gives $I_Z = (9.6 - 5.1) / 1.2 \text{ k}\Omega = 3.75 \text{ mA}$, which keeps the Zener in regulation without an excessive parasitic draw on the pack.

B. Voltage Sensing Divider (R2, RV1, R3)

The comparator trips when the sensed voltage falls below the 5.1 V reference. With R_{top} the series combination of R2 and RV1, and R3 the bottom leg to ground,

$$V_{\text{sense}} = V_{\text{rail}} \times [R_3 / (R_{\text{top}} + R_3)] \quad (2)$$

Solving for the top resistance required to trip the comparator at exactly 9.6 V, with $V_{\text{ref}} = 5.1 \text{ V}$ and $R_3 = 10 \text{ k}\Omega$,

Designator	Value / Part No.	Component	Function
U1:A	LM324	Op-amp (1 of 4)	Voltage comparator
B1	3S Li-ion, 12.6–9.6 V	Battery pack	Energy source
—	LM2596	Buck converter	9 V regulated output

$$R_{\text{top}} = R_3 \times (V_{\text{ref}} / V_{\text{cutoff}} - 1) = 10 \text{ k}\Omega \times (5.1/9.6 - 1) = 8.823 \text{ k}\Omega \quad (3)$$

This target is realized with an 8.2 k Ω fixed resistor (R2) in series with the wiper of a 10 k Ω precision trimpot (RV1), calibrated so that the trimpot contributes $\approx 623 \Omega$, giving $R_{\text{top}} \approx 8.823 \text{ k}\Omega$ as required.

C. BJT Base Current Limiting (R4)

When the LM324 output swings HIGH ($\approx 11.1 \text{ V}$ at full charge), the base current delivered to Q1 must be limited to avoid exceeding the transistor's and the op-amp's ratings:

$$I_B \approx (V_{\text{out}} - 0.7 \text{ V}) / R_4 \quad (4)$$

With $R_4 = 4.7 \text{ k}\Omega$ the base current is capped at approximately 1.57 mA — sufficient to saturate Q1 while remaining well within the LM324's maximum source-current rating.

D. MOSFET Gate Discharge and Conduction Loss (R5)

A 10 k Ω pull-down resistor (R5) ties the IRFZ44N gate to ground so that when the comparator output goes LOW the gate cannot float; it rapidly discharges the gate capacitance and forces $V_{\text{gate(OFF)}} = 0 \text{ V}$, giving an instantaneous load cutoff. While the MOSFET is ON, its conduction loss is

$$P_{Q2} = I_{\text{load}}^2 \times R_{\text{DS(on)}} \quad (5)$$

For a load current of $\approx 1 \text{ A}$ and the IRFZ44N's $R_{\text{DS(on)}}$ of 28 m Ω , this loss is under 30 mW,

confirming the MOSFET is an appropriate low-loss ground-path switch for this application.

VI. CIRCUIT OPERATION

Normal operation ($V_{\text{batt}} > 9.6 \text{ V}$): the divided battery voltage at the LM324's non-inverting input exceeds the 5.1 V reference at the inverting input, so the comparator output is HIGH. Q1 saturates, pulling the MOSFET gate HIGH, and Q2 conducts, completing the ground path for the buck converter and the router.

Buck-converter dropout zone ($10.5 \text{ V} > V_{\text{batt}} > 9.6 \text{ V}$): once the battery voltage falls below 10.5 V, the LM2596 exhausts its $\approx 1.5 \text{ V}$ dropout headroom and its output sags below 9 V. The router tolerates this minor sag through its internal 5 V/3.3 V logic regulators.

Low-voltage disconnect ($V_{\text{batt}} \leq 9.6 \text{ V}$): the sensed voltage drops below 5.1 V, the LM324 output snaps LOW, Q1 cuts off, R5 forces the MOSFET gate to 0 V, and the ground path is severed. The

VIII. RESULTS AND DISCUSSION

The complete circuit was simulated over a 100 s window in which mains power is removed at $t = 40 \text{ s}$ (Fig. 2). Prior to the outage the pack floats at 13.8 V while the buck converter supplies a stable load current of $\approx 1.35 \text{ A}$. At the moment of grid loss the transfer to battery power is seamless, with no interruption visible in the load current trace.

As the battery discharges from 12.1 V toward the 10.5 V buck-converter dropout point, the load current declines gradually from $\approx 1.2 \text{ A}$ to $\approx 1.05 \text{ A}$. Once V_{batt} reaches the 9.6 V hardware threshold, at $t = 64.3 \text{ s}$ (i.e., 24.3 s after the outage began), the comparator output (V_{gate}) collapses from $\approx 4 \text{ V}$ to 0 V and the IRFZ44N turns off within milliseconds, dropping the load current from $\approx 1.05 \text{ A}$ to near 0 A. The battery voltage then plateaus just at the cutoff line, confirming that the pack is protected from further discharge into the deep-discharge region.

system then draws near-zero current until grid power is restored and the pack is recharged.

VII. HARDWARE IMPLEMENTATION AND TROUBLESHOOTING

Initial breadboard testing revealed erratic switching and extreme touch-sensitivity at the BC547 base: the load would intermittently re-energize at low battery voltage whenever the logic-stage wiring was touched. Diagnosis traced this to a floating-node condition caused by a split-rail discontinuity on the breadboard — the LM324 ground (pin 11) and the Zener reference were physically isolated from the main supply ground, forcing the op-amp into a high-impedance, undefined output state. Bridging the upper and lower ground rails established a single common reference potential, allowing the LM324's push-pull output stage to actively drive the logic LOW, which permanently resolved the parasitic-capacitance interference.

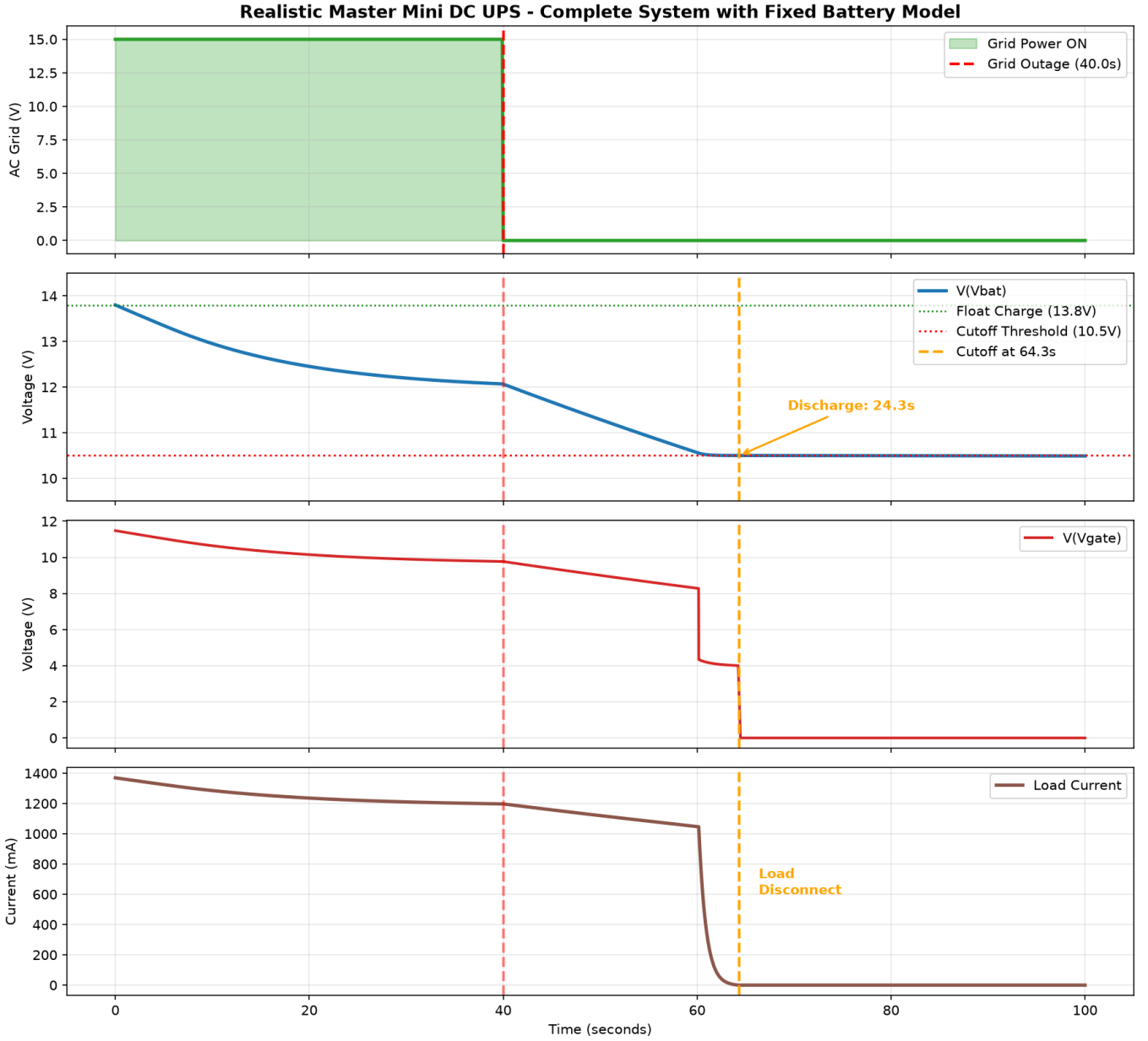


Fig. 2. Simulated grid-outage response: AC grid status, battery voltage, comparator gate-drive voltage, and load current versus time.

These results confirm that the analog comparator scheme achieves the three design objectives of Section III: (i) uninterrupted, glitch-free transfer to battery backup, (ii) a regulated 9 V output tolerant of the buck converter's dropout region, and (iii) a hard, deterministic disconnect that prevents the 3S Li-ion pack from being driven below its 9.6 V safe limit, without requiring any microcontroller or firmware.

IX. CONCLUSION

A fully analog 3S Li-ion DC UPS with hardware-enforced deep-discharge protection has been designed, mathematically modeled, and validated in

Proteus simulation. The circuit combines a Zener voltage reference, an LM324 comparator, and a BC547/IRFZ44N driver-switch pair to deterministically disconnect the load at a calibrated 9.6 V threshold, reserving 10–15% of pack capacity and thereby extending cell life, while an LM2596 buck stage maintains a regulated 9 V rail for the protected router. A hardware grounding fault discovered during breadboard testing — and its resolution by unifying the split ground rails — underscores the practical importance of a common reference potential in open-loop comparator designs. Future work will add switching hysteresis around the

9.6 V threshold to eliminate any risk of relay-like chatter near the trip point, incorporate over-temperature protection, and explore a low-power digital state-of-charge indicator that supplements, rather than replaces, the analog safety cutoff.

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